EMILIE (LIUYING) CHEN

Computational Science | Optimization | Machine Learning

London, UK | chenliuying666@163.com | github.com/ada-lc826

EDUCATION

Imperial College London

MSc Applied Computational Science and Engineering | Sep 2026 - Nov 2027 (expected)

Computational mathematics, numerical programming, data science and machine learning.

Shanghai University - Sino-European School of Technology

BEng Information Engineering | Sep 2022 - Jun 2026

Shanghai Outstanding Graduate. GPA: 3.33/4.0 (85.28/100).

RESEARCH & RECOGNITION

MCM / ICM 2025 - Finalist Award

Mathematical modelling | Lead paper writer

Sustainable tourism modelling with NSGA-II multi-objective optimization, Sobol sensitivity analysis, and Monte Carlo simulation.

Research on Production Decision Making Based on Genetic Algorithm

Co-first author | Liuying Chen, Yi Chen, Jiahao Wang (equal contribution)

Genetic algorithms, 0-1 integer programming and dynamic programming for inspection and disassembly decisions in multi-stage production.

Tilting-Rotor Quadrotor Trajectory Tracking and Attitude-Constrained Control

Co-author | IEEE ICUS 2025

Linear model predictive control for trajectory tracking with attitude constraints.

Other recognition: APMCM 2024 Third Prize; software copyright co-holder for a production process optimization and decision management system.

EXPERIENCE

Algorithm Intern - Vconics Automotive

Jun - Jul 2025 | Shanghai

Worked on model predictive control, state estimation and automotive engineering using MATLAB/Simulink.

Data Analysis Intern - Polestar

Aug - Sep 2023 | Shanghai

SQL analytics, BERT / BiLSTM-CRF NLP workflows, and business intelligence dashboards.

SELECTED PROJECTS & LEADERSHIP

MediChina - Multilingual medical-travel concept platform. React, TypeScript, Tailwind CSS.
GuardianWing - Smart-walker and AI-care interactive concept. HTML, CSS, JavaScript.
Paper Isle - Reading map for bookstores and libraries, with private local check-ins.
FingerFight - Canvas typing game with word battles, racing and performance feedback.
Student Union President, Shanghai University - Led a 70-person team.

SKILLS & LANGUAGES

Python / PyTorch / NumPy; C / C++; SQL; MATLAB / Simulink; Git; LaTeX.
Mandarin (native); English (working proficiency); French (TCF B1).